MASTER'S COMPREHENSIVE EXAMINATION

Part I **Theory (90 minutes)** **March 18, 2006**

Instructions: Answer all four questions. Open book and open notes.

1. A remote beacon contains two batteries, a primary and a spare. The beacon runs on the primary battery until it fails, at which point it starts using the spare. When both batteries fail, the beacon stops operating. The lifetimes (in months) of the batteries are independent exponential random variables with pdf $f(x) = \lambda e^{-\lambda x}$, $x > 0$, $\lambda > 0$. Find the probability the beacon will stop operating in less than T months.

2. The prevalence of a certain disease in a particular population is 1 in 100. A blood test can detect the disease. To save time, tests are performed in batches of 10 people. Such a pooled test is negative only if no one among the 10 being tested had the disease.

(a) Mr. A is tested in this way and the pooled test result is positive. What is the probability Mr. A has the disease?

(b) Mr. A is retested with 9 different people (than before) and that pooled test also comes back positive. Now what is the probability Mr. A has the disease?

3. Suppose X is an observation from the distribution with pdf

$$f(x|\theta) = \begin{cases} 2\theta^2/x^3, & x > \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the MLE of θ.

(b) Find a 90% Confidence interval for θ of the form $aX < \theta < bX$. That is, choose a and b so that $\mathrm{P}(aX < \theta < bX) = 0.9$.

4. A quality control engineer examines blank magnetic disks successively until finding one with a defect. Let X be the number of disks examined (including the defective one) and let p be the proportion of disks that actually have defects. To test $\mathrm{H}_0 : p = 1/10$ *vs* $\mathrm{H}_1 : p < 1/10$, reject H_0 if $X \geq 28$.

(a) Find the size (say α) of this test.

(b) Find the type II error (say β) of the test when $p = .05$.

(c) If $X = 30$, find the p-value for X.